Maximum Crown Width Prediction Equation

Greg Johnson, Aaron Weiskittel, David Marshall

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Maximum Crown Width (`MCW`) Prediction

Data

We used the Forest Health Monitoring data supplied by Aaron. It contains 127710 observations on 221 species. Earlier analysis indicated that trees with crown ratios (`cr`) below 0.5 were suspect for determining maximum crown width. We filtered the data to exclude low crown ratios (`crs`), yielding 71255 observations on 215 species. Additionally, we computed the sample size by species and filtered species with less than 200 observations, yielding 64328 observations on 65 species.

Aaron and David previously determined that the 95th percentile of crown width (`cw`) was a useful approximation for maximum crown width (`mcw`).

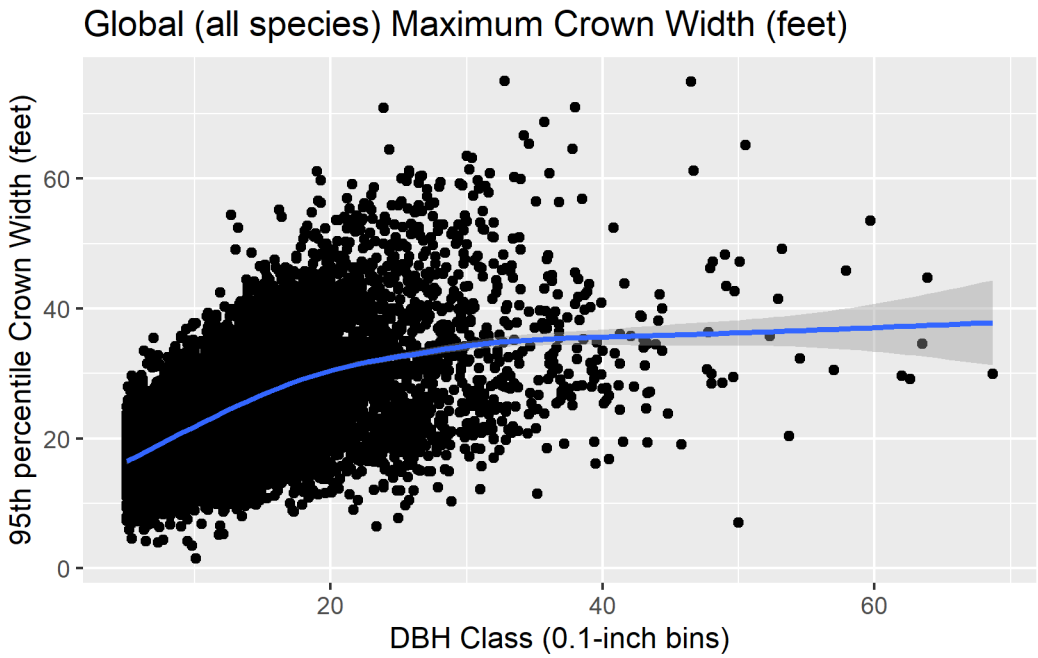
We split the analysis in two phases:

1. Develop equation form (system) and initial parameter estimates by computing the 95th percentile `cw` for 0.1 inch diameter at breast height (`dbh`) classes for each species.
2. Fit the final model with quantile regression on the filtered data.

Phase 1

Global Classed `mcw`

Below is a graph of `mcw` (approximated using species-specific 0.1-inch `dbh` classes' 95th percentile `cw`). The graph illustrates that over all species, crown width tends toward an asymptote.



We fit the filtered and classed data using Equation 1:

$$cw_{95} = \beta_0 dbhc^{\beta_1} \tag{1}$$

where:

- cw_{95} - 95th percentile crown width (feet) for species-specific 0.1-inch `dbh` class
- $dbhc$ - 0.1-inch `dbh` class (inches)

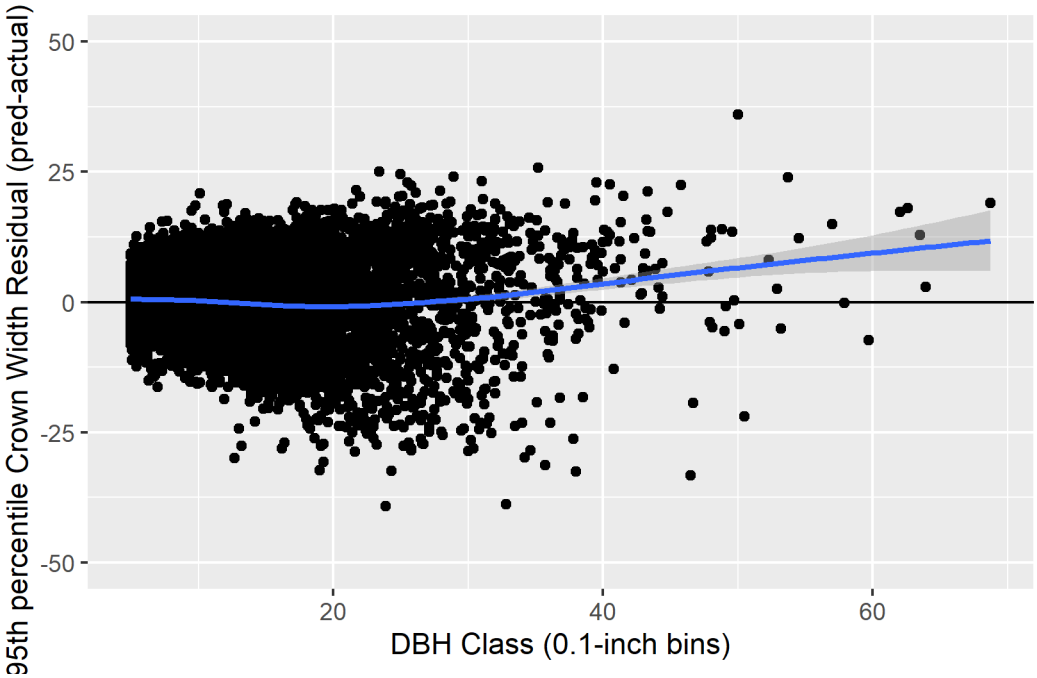
The fit statistics are shown below with a residual graph. The global fit does not show serious residual trends across `dbh`, but there is a significant difference between conifers and hardwoods.

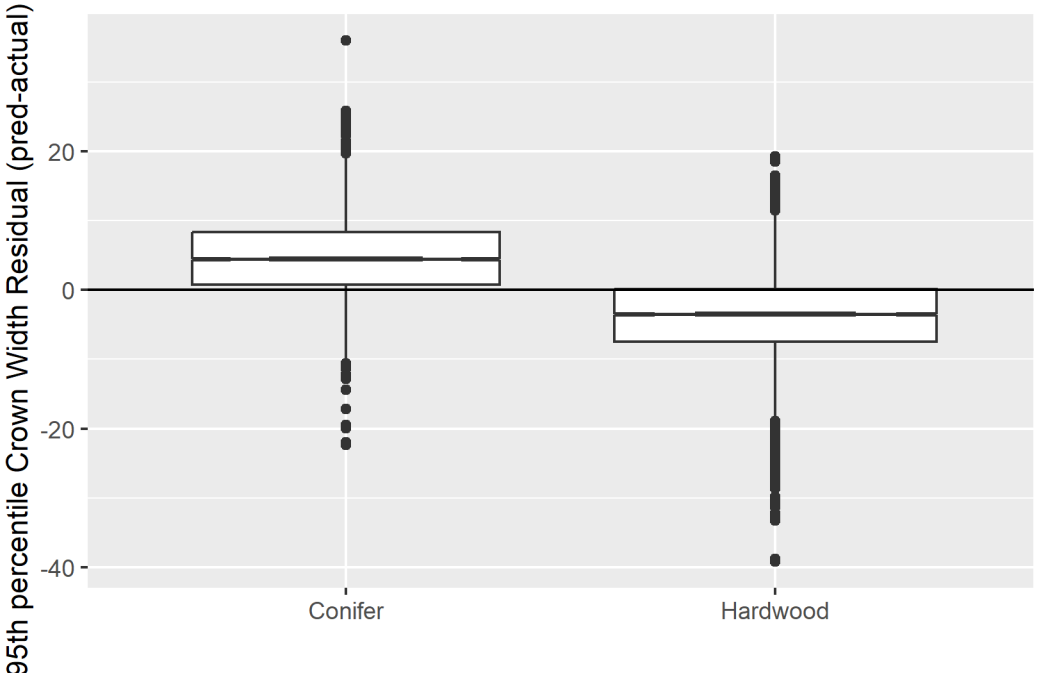
```
Formula: cw_95 ~ b0 * dbh_class^b1

Parameters:
   Estimate Std. Error t value Pr(>|t|)
b0 8.602544   0.151220   56.89   <2e-16 ***
b1 0.411441   0.006393   64.36   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 7.504 on 8875 degrees of freedom

Number of iterations to convergence: 7
Achieved convergence tolerance: 4.997e-06
```





Global Classed `mcw` with Conifer/Hardwood Differentiation

We adjusted Equation 1 to add an indicator variable (`type` (1 if conifer, 0 if hardwood) to adjust β_0 and β_1. We found that the β_0 adjustment was not significant, so the resulting new equation is Equation 2:

$$cw_{95} = \beta_0 dbhc^{(\beta_1 + \beta_{1a} type)} \tag{2}$$

where:

- cw_{95} - 95th percentile crown width (feet) for species-specific 0.1-inch `dbh` class
- $dbhc$ - 0.1-inch `dbh` class (inches)
- $type$ - 1 if conifer, 0 if hardwood

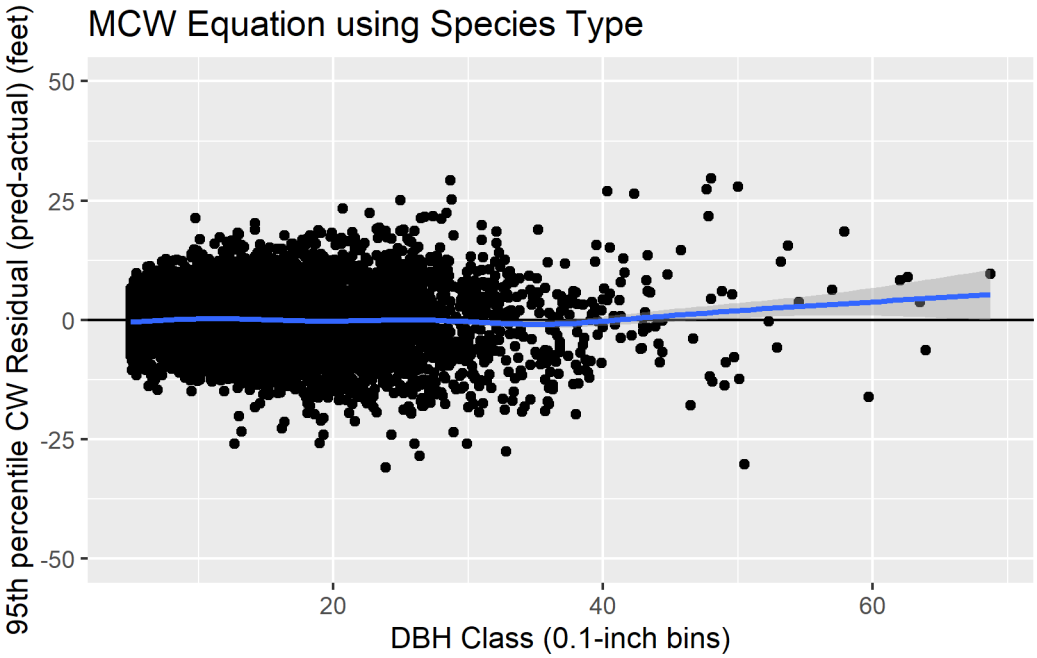
The fit statistics for Equation 2 are shown below with a residual graph. The new global fit does not show serious residual trends across `dbh`, and removes the difference between conifers and hardwoods. Looking at residuals at the species level reveals that there are species-level effects that need to be accounted for (see last graph in this section).

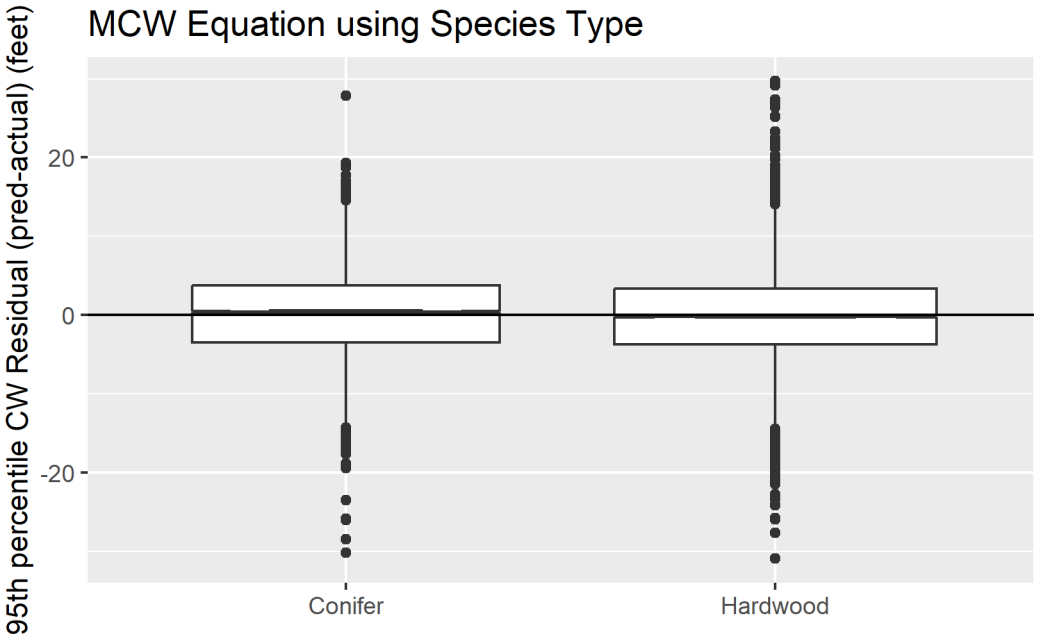
```
Formula: cw_95 ~ (b0) * dbh_class^(b1 + b1a * (type == "Conifer"))

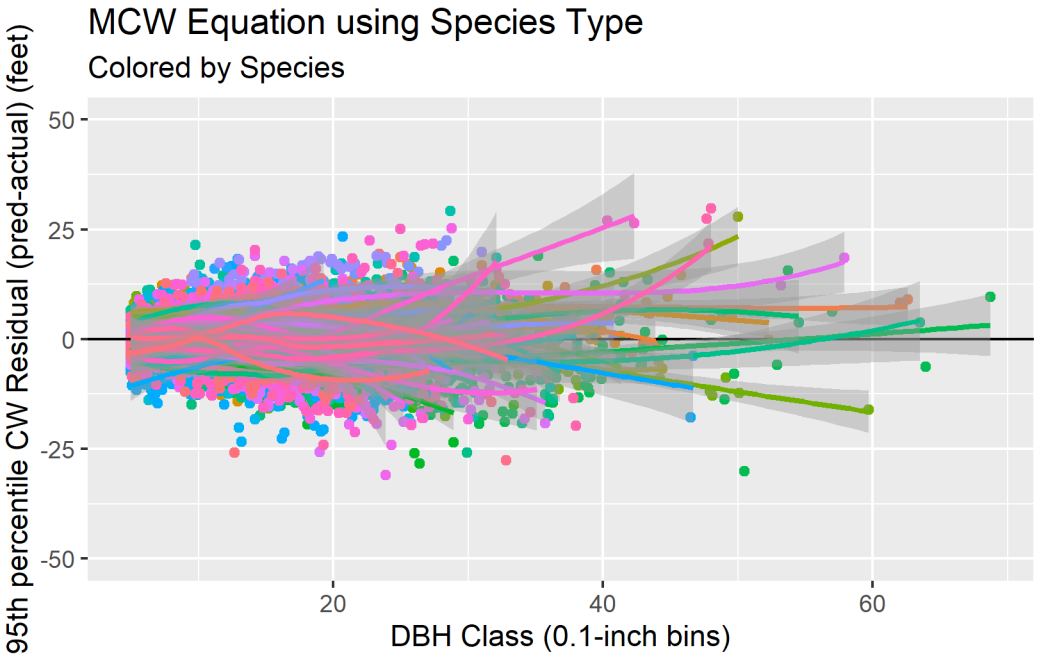
Parameters:
     Estimate Std. Error t value Pr(>|t|)
b0   7.324708   0.103802   70.56   <2e-16 ***
b1   0.535148   0.005264  101.66   <2e-16 ***
b1a -0.136165   0.001883  -72.33   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.894 on 8874 degrees of freedom

Number of iterations to convergence: 3
Achieved convergence tolerance: 1.49e-08
```

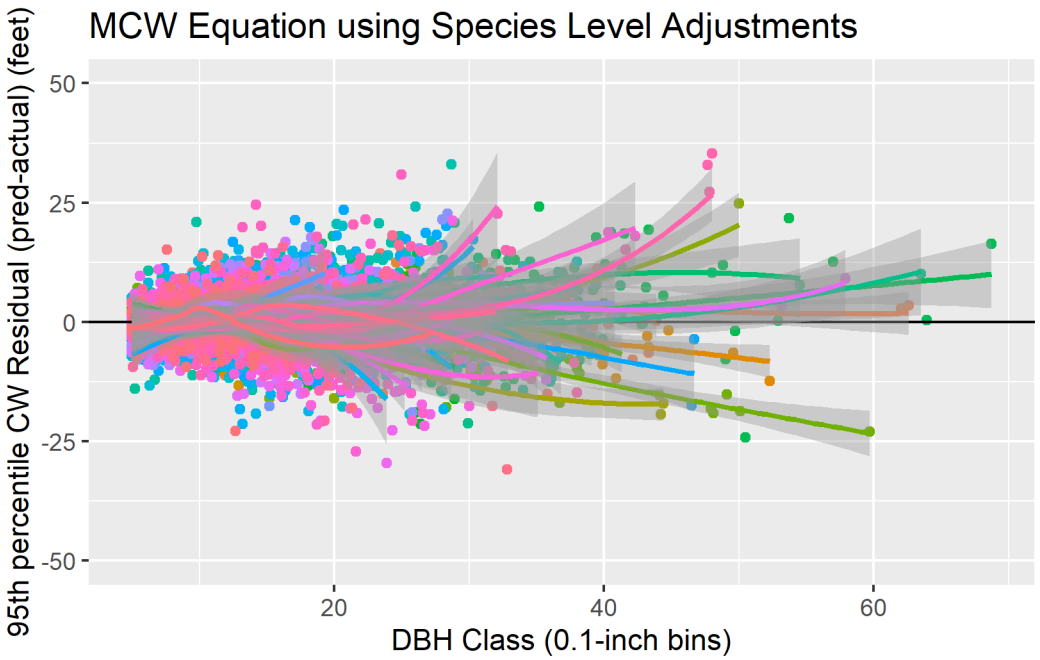

MCW Equation using Species Type
95th percentile CW Residual (pred-actual) (feet)
50
25
0
-25
-50
20
40
60
DBH Class (0.1-inch bins)

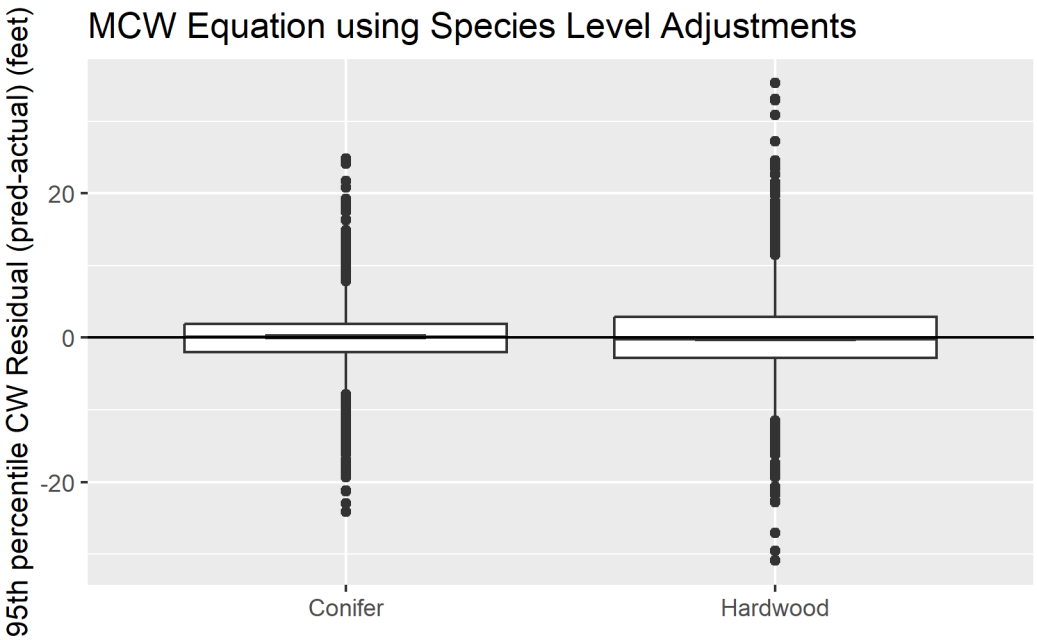

MCW Equation using Species Type
95th percentile CW Residual (pred-actual) (feet)
20
0
-20
Conifer
Hardwood


MCW Equation using Species Type
Colored by Species
95th percentile CW Residual (pred-actual) (feet)
50
25
0
-25
-50
20
40
60
DBH Class (0.1-inch bins)

Global Classed `mcw` with Conifer/Hardwood Differentiation and Species-specific Adjustments

To maintain the global curve shape and stable estimates provided by Equation 2, we computed the mean ratio of Equation 2 predictions to cw_{95} by species on the 0.1-inch `dbh` classed data. The graphs below show the residuals after the adjustment. The adjustments reduce the species-specific biases.




MCW Equation using Species Level Adjustments
95th percentile CW Residual (pred-actual) (feet)
20
0
-20
Conifer
Hardwood

Phase 2

Parameter Estimates using Quantile Regression

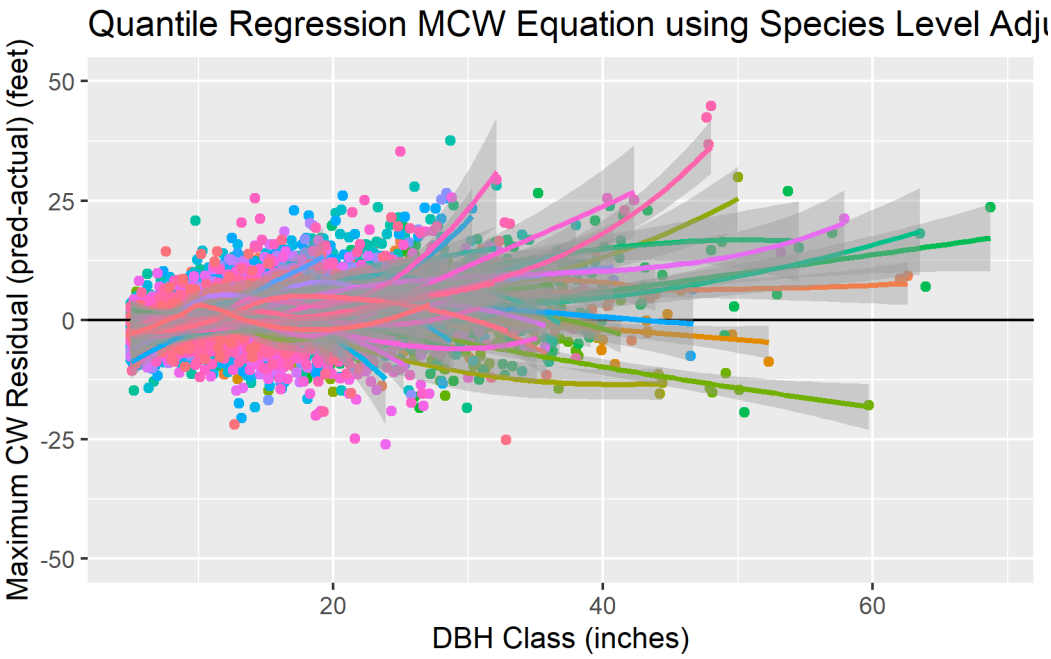
Using the filtered `cw` data we fit Equation 2 to the individual tree data using quantile regression with a $\tau = 0.95$. The predictions from the quantile regression were then used to compute species-level adjustment ratios. Those ratios were applied to the tree-level data.

The regression summary statistics for the quantile regression are shown below along with a table showing the adjusted β_0 parameter estimates by species.

```
Call: quantreg::nlrq(formula = cw_ft ~ b0 * dbh_in^(b1 + b1a * (type ==
    "Conifer")), data = crown_width_data2, start = c(b0 = 5.3247,
    b1 = 0.435, b1a = -0.1362), tau = 0.95, trace = FALSE, control = list(
    maxiter = 100, k = 2, InitialStepSize = 1, big = 1e+20, eps = 1e-07,
    beta = 0.97))

tau: [1] 0.95

Coefficients:
    Value     Std. Error t value   Pr(>|t|)
b0    6.91138   0.07266   95.11370   0.00000
b1    0.64847   0.00505  128.33918   0.00000
b1a  -0.13670   0.00183  -74.75786   0.00000
```


Quantile Regression MCW Equation using Species Level Adju
Maximum CW Residual (pred-actual) (feet)
50
25
0
-25
-50
20
40
60
DBH Class (inches)

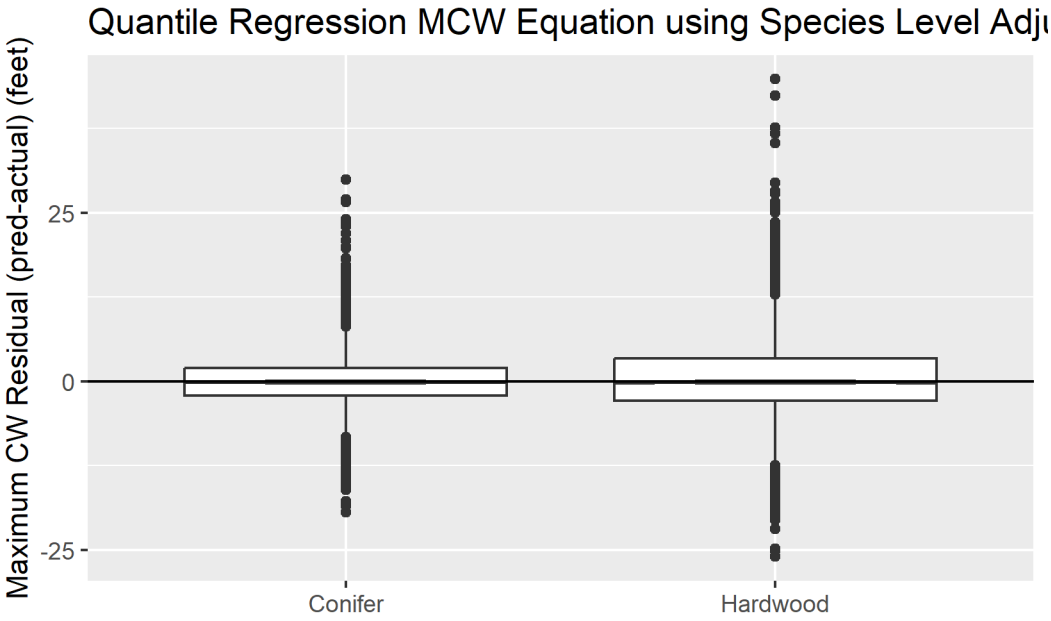

Quantile Regression MCW Equation using Species Level Adju
Maximum CW Residual (pred-actual) (feet)
25
0
-25
Conifer
Hardwood

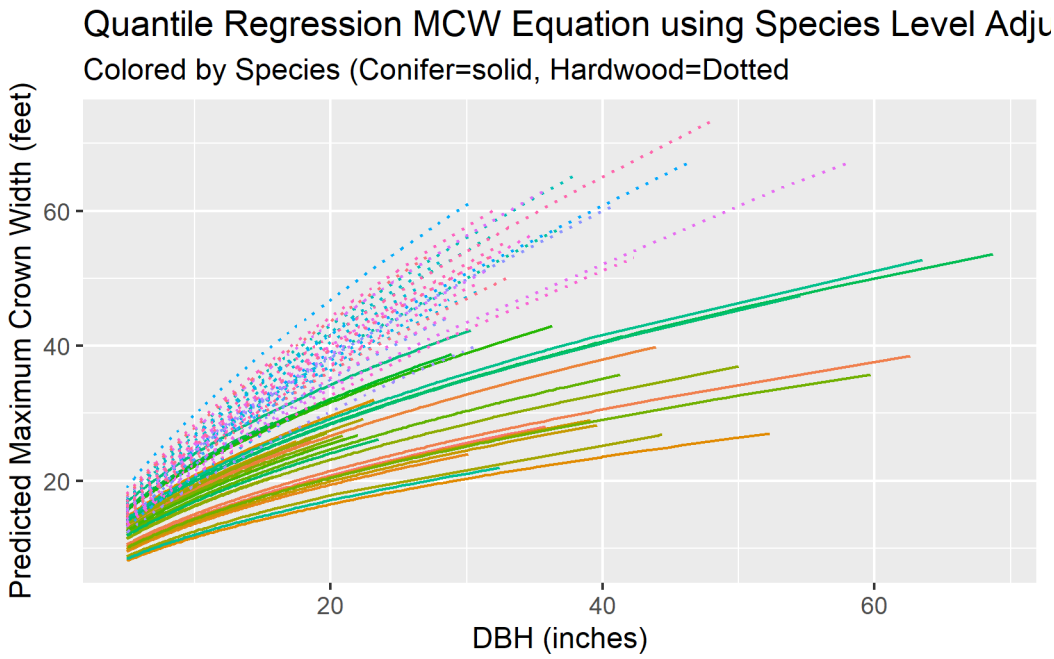
Table 1: Adjusted b0 Parameters

spcd	adjustment ratio	b0_adj
11	0.6497287	4.490522
12	0.8379334	5.791276
15	0.6701136	4.631409
17	0.8327396	5.755379
19	0.6065698	4.192234
20	0.5160281	3.566466
68	0.9264391	6.402972
71	0.8653523	5.980778
81	0.6423221	4.439332
93	0.6219067	4.298233
94	0.7985177	5.518859
95	0.7273333	5.026876
97	0.8587229	5.934959
101	0.5570410	3.849921
105	0.8529231	5.894875
108	0.7224179	4.992904
111	0.8157672	5.638077
116	0.6376444	4.407002
122	0.7701865	5.323051
125	0.7958140	5.500172
129	0.9882569	6.830218
131	1.0024077	6.928020
132	1.0093763	6.976183
202	0.8902039	6.152537
241	0.7521579	5.198449
242	0.8867202	6.128459
261	1.0689995	7.388261
263	0.9117819	6.301670
264	0.5344061	3.693483
313	0.8005228	5.532717
316	0.8628707	5.963627
318	0.8935591	6.175726
351	0.7920041	5.473840
371	0.8725800	6.030732
372	0.8411958	5.813823
375	0.7921412	5.474788
400	0.7539459	5.210806
403	0.7849524	5.425103
407	0.7685349	5.311636

spcd	adjustment ratio	b0_adj
409	0.8648374	5.977219
531	0.9707103	6.708947
541	0.8086558	5.588927
543	0.6720723	4.644947
591	0.7555304	5.221757
611	0.7263159	5.019844
621	0.7941411	5.488611
631	0.6282711	4.342220
693	0.8177446	5.651743
711	0.8053614	5.566158
746	0.7081075	4.893999
762	0.7915671	5.470820
802	0.8972232	6.201050
805	0.6976343	4.821615
806	0.8296737	5.734190
807	0.7304337	5.048305
812	0.8185528	5.657329
818	0.6771569	4.680088
823	0.7730050	5.342530
827	0.9191050	6.352283
832	0.8323745	5.752856
833	0.8610400	5.950974
835	0.7677773	5.306400
837	0.8187864	5.658943
951	0.7504575	5.186696
972	0.9066190	6.265988

Equation Performance

Below is a graph of equation predictions by species.



Conclusion

The system developed above accounts for species-level variation in `cw` without sacrificing biologically meaningful equation performance. It is unbiased against the fitting data. Generally, hardwoods have wider crowns than conifers.

The system does produce some high predicted crown widths that should be investigated.